

Automating Bloom's Taxonomy Classification of Educational Content: A GPT-4-Based Approach

Background

Bloom's taxonomy is a fundamental framework for categorizing educational objectives and assessment items into levels of cognitive complexity (e.g., Remember, Understand, Apply, Analyze, Evaluate, Create). Identifying the Bloom's level of questions or learning outcomes is important for curriculum design and assessment alignment, but doing so typically relies on manual expert judgment, which can be time-consuming and inconsistent ¹. Recent advances in **large language models (LLMs)**, especially GPT-4, offer a potential solution by automatically interpreting and classifying educational content. By leveraging GPT-4's natural language understanding, we may automate the classification of exam questions, learning outcomes, and other instructional text according to Bloom's taxonomy. The research problem is to determine how effectively an LLM like GPT-4 can perform this cognitive level classification and how to optimize its performance, ensuring the automated classifications are accurate, reliable, and aligned with human expectations.

Research Objectives

- **Objective 1: Develop an LLM-based classification system** – To develop a prototype method using a large language model (e.g., GPT-4) that automatically classifies educational content (such as exam questions or learning outcome statements) into the correct Bloom's taxonomy level, and to evaluate its performance using metrics like classification accuracy, precision/recall, and Cohen's κ (inter-rater agreement).
- **Objective 2: Optimize prompting and model use** – To investigate how different strategies for using the LLM (zero-shot vs. few-shot prompting, prompt engineering techniques, or fine-tuning on a labeled dataset) affect the classification accuracy and consistency, identifying which approach yields the most reliable Bloom's level categorization.
- **Objective 3: Benchmark against humans and baselines** – To compare the LLM-based classifications with those by human experts and with traditional automated methods (e.g., rule-based keyword classifiers or fine-tuned non-LLM models) in order to assess the model's validity, understand its strengths/weaknesses, and quantify how closely the AI system can replicate expert classification of cognitive levels.
- **Objective 4: Explore practical integration** – To explore the integration of the developed Bloom's classification tool into an educational technology context (for instance, as a feature in e-learning platforms or teacher support tools), and to analyze its potential impact on educational practice along with any challenges (such as interpretability of results, required prompt refinements, or handling of ambiguous cases) that arise in real-world use.

Research Questions

- **RQ1:** *How accurately and consistently can a GPT-4-based large language model classify educational content into the correct Bloom's taxonomy categories, and how does its performance compare to human educators' classifications?*

- **RQ2:** Which utilization approach for an LLM yields the best results for Bloom's taxonomy classification – for example, does carefully engineered zero-shot/few-shot prompting outperform fine-tuning a model on labeled data in terms of classification accuracy and agreement with human labels?
- **RQ3:** How does the LLM-driven classification compare with traditional automated methods and expert judgments? In particular, what are the common areas of discrepancy or confusion (e.g., between adjacent Bloom's levels like Remember vs. Understand), and what do these reveal about the model's strengths and limitations?
- **RQ4:** What are the practical implications of deploying an LLM-based Bloom's classification tool in educational settings? For instance, how might such a tool support instructors in analyzing or designing questions and learning outcomes, and what challenges (such as trust, usability, or domain-generalization issues) need to be addressed for effective real-world adoption?

Addressing Literature Gaps and Trends

These objectives and questions are designed to fill a clear gap in the intersection of educational technology and AI. While initial studies have shown that GPT-4 can classify learning outcomes or teacher questions with moderate success (achieving around 60–65% accuracy and moderate agreement with experts) ² ³, the area is still nascent. By systematically evaluating *how* to best use LLMs for Bloom's taxonomy classification (and not just whether they can), this research addresses the need for deeper understanding and improvement of these early results. It aligns with current trends in **educational NLP and AI**, where there is growing interest in applying generative models to assist with pedagogical design, assessment, and learning analytics. By focusing on automated cognitive level classification, the study contributes to modern efforts in AI-driven curriculum development and assessment tools, helping to bridge the gap between advanced NLP capabilities and practical needs in education. The outcomes will shed light on the feasibility of using state-of-the-art LLMs to perform nuanced educational analyses, thereby extending the literature on AI in education and providing guidance for educators and developers looking to integrate such technologies into educational platforms.

¹ ³ (PDF) Automated Bloom's Taxonomy Classification of Teacher Questions Using Whisper and GPT-4

https://www.researchgate.net/publication/399287377_Automated_Bloom's_Taxonomy_Classification_of_Teacher_Questions_Using_Whisper_and_GPT-4

² Leveraging generative AI for course learning outcome categorization using Bloom's taxonomy - ScienceDirect

<https://www.sciencedirect.com/science/article/pii/S2666920X2500044X>